

Econ 8400-001, Fall 2025
Problem Set 4 (Due November 20)

In Rust's (1987) problem, assume:

$$c(x_t; \theta) = \theta_1 x + \theta_2 x^2$$

$$x_{t+1} = \begin{cases} x_t + 1 & \text{with Pr} = 1 - \theta_3 \\ x_t & \text{with Pr} = \theta_3 \end{cases}$$

and that the replacement cost is θ_4 .

Assume particular values for the parameters.

1. Solve the problem (as in Rust 87) discretizing x into $K=90$ ($x = 0 : 1 : 89$) values and iterating over $V_\sigma(x) \equiv EV(x)$.
2. Solve the problem through the fixed point of probabilities $P(a|x)$ (i.e., policy function iteration) still using $k=90$. Compute $V_\sigma(x)$ implied by the probabilities you found. Make sure your solution coincides with the solution in (1).
3. Build a likelihood based on the probabilities you found above that can be used to estimate θ_1 , θ_2 , and θ_4 .
4. Using the parameters you assumed, simulate 10000 observations (i.e., pairs of mileage and replacement). Show how would you use them to estimate the parameters using Hotz and Miller (1993).
5. Optional: Use the data from (4) (or a subset of it) and the likelihood from (3) to estimate θ_1 , θ_2 , and θ_4 . How do the estimates compare to the values you used to simulate the data?